



Experiment 6

Prepared By:

Name: Mahin Raj

Course: BCA (2nd Semester)

SAP ID: 590025881

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Submitted To:

Mr. Vibhu Gautam

Faculty, School of Computer Science

UPES, Dehradun



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mahin-mr12 commit

9eed2b6 · 3 weeks ago



326 lines (326 loc) · 8 KB

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Experiment 6: Lists, tuples, dictionary

1. Scan n values in range 0-3 and print the number of times each value has occurred.

```
In [ ]: n = int(input("Enter number of values: "))

values = []
for i in range(n):
    num = int(input("Enter value (0-3): "))
    if 0 <= num <= 3:
        values.append(num)
    else:
        print("Invalid input! Enter between 0 and 3")

# Count occurrences
for i in range(4):
    print(f"{i} occurred {values.count(i)} times")
```

```
0 occurred 0 times
1 occurred 1 times
2 occurred 1 times
3 occurred 0 times
```

2. Create a tuple to store n numeric values and find the average of all values.

```
In [ ]: n = int(input("Enter number of elements: "))

nums = []
for i in range(n):
    nums.append(float(input("Enter number: ")))

t = tuple(nums)

average = sum(t) / len(t)

print("Tuple:", t)
print("Average:", average)
```

Tuple: (45.0, 42.0, 43.0, 46.0)

Average: 44.0

3. WAP to input a list of scores for N students in a list data type. Find the score of the runner-up and print the output.

Sample Input

N = 5

Scores= 2 3 6 6 5

Sample output

5

Note: Given list is [2, 3, 6, 6, 5]. The maximum score is 6, second maximum is

5. Hence, we print 5 as the runner-up score.

```
In [ ]: N = int(input("Enter number of students: "))
scores = list(map(int, input("Enter scores separated by space: ").split()))

unique_scores = list(set(scores)) # remove duplicates
unique_scores.sort()

runner_up = unique_scores[-2]

print("Runner-up score:", runner_up)
```

Runner-up score: 6

4. Create a dictionary of n persons where key is name and value is city.

- Display all names
- Display all city names
- Display student names and cities of all students.
- Count the number of students in each city.

```
In [ ]: n = int(input("Enter number of students: "))
```

```

students = {}

for i in range(n):
    name = input("Enter name: ")
    city = input("Enter city: ")
    students[name] = city

# a. Display all names
print("Names:", students.keys())

# b. Display all cities
print("Cities:", students.values())

# c. Display all student details
print("Student Details:")
for name, city in students.items():
    print(name, "lives in", city)

# d. Count students in each city
city_count = {}
for city in students.values():
    city_count[city] = city_count.get(city, 0) + 1

print("Students in each city:", city_count)

```

```

Names: dict_keys(['krish', 'raj'])
Cities: dict_values(['patna', 'raipur'])
Student Details:
krish lives in patna
raj lives in raipur
Students in each city: {'patna': 1, 'raipur': 1}

```

5. Store details of n movies in a dictionary by taking input from the user. Each movie must store details like name, year, director name, production cost, collection made (earning) & perform the following :-
 - a. print all movie details
 - b. display name of movies released before 2015
 - c. print movies that made a profit. d. print movies directed by a particular director.

```

In [ ]: n = int(input("Enter number of movies: "))
        movies = {}

        for i in range(n):

```

```

name = input("Movie name: ")
year = int(input("Release year: "))
director = input("Director name: ")
cost = float(input("Production cost: "))
collection = float(input("Collection made: "))

movies[name] = {
    "year": year,
    "director": director,
    "cost": cost,
    "collection": collection
}

# a. Print all movie details
print("\nAll Movie Details:")
for name, details in movies.items():
    print(name, details)

# b. Movies released before 2015
print("\nMovies released before 2015:")
for name, details in movies.items():
    if details["year"] < 2015:
        print(name)

# c. Movies that made profit
print("\nProfitable Movies:")
for name, details in movies.items():
    if details["collection"] > details["cost"]:
        print(name)

# d. Movies by particular director
search_director = input("\nEnter director name to search: ")
print("Movies directed by", search_director)
for name, details in movies.items():
    if details["director"].lower() == search_director.lower():
        print(name)

```

All Movie Details:

```

rrr {'year': 2024, 'director': 'raja moli', 'cost': 500.0, 'collection': 900.0}
kick {'year': 2018, 'director': 'unknow', 'cost': 350.0, 'collection': 600.0}

```

Movies released before 2015:

Profitable Movies:

```

rrr
...

```

```
KICK
```

```
Movies directed by raja moli
```

```
rrr
```